

Supplemental Material for Response to Comment on “Posterior comparison of model dynamics in several hybrid turbulence model forms” [Phys. Fluids 36, 105148 (2024)]

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The following supplemental materials compliment our Response to the Comment by Heinz¹ on our article Towery, Sáenz, and Livescu² (herein referred to as TSL24). Please see TSL24 for complete definitions of all terms and mathematical symbols presented in this supplement.

In Fig. 1 we present the root-mean-square errors (RMSE) at each filter wavenumber κ_f for the kinetic energy (KE) and KE dissipation resolution fractions computed from each time series as follows. First, all spatially-averaged time series of $\mathcal{K}_r$, $\mathcal{K}_m$, $\mathcal{E}_r$, and $\mathcal{E}_m$, from the filtered DNS, LES, SKE, and HRL are interpolated to a set of $N_t = 100$ uniformly-spaced time points using cubic splines. Then the resolution fractions F_k and F_ε for each model and DNS were computed at these points and the RMSE was computed using the standard formula

$$\text{RMSE}(F_x) = \left(\frac{1}{N_t} \sum_{n=1}^{N_t} [F_x(t^n) - F_x^{\text{dns}}(t^n)]^2 \right)^{1/2} \quad \text{where } t^{n+1} - t^n = t_{\text{end}}/(N_t - 1) \forall t^n \in [0, t_{\text{end}}]. \quad (1)$$

It can be seen that RMSE of F_k and $1 - F_k$ are identical, for the obvious mathematical reason, however we show both anyway to compare with Figs. 2 and 3. Furthermore, it can be seen that no single model has the lowest RMSE in F_k across all κ_f , but the CES-S models are best or competitive from $\kappa_f = 8$ to 64. In contrast, the CES-S models

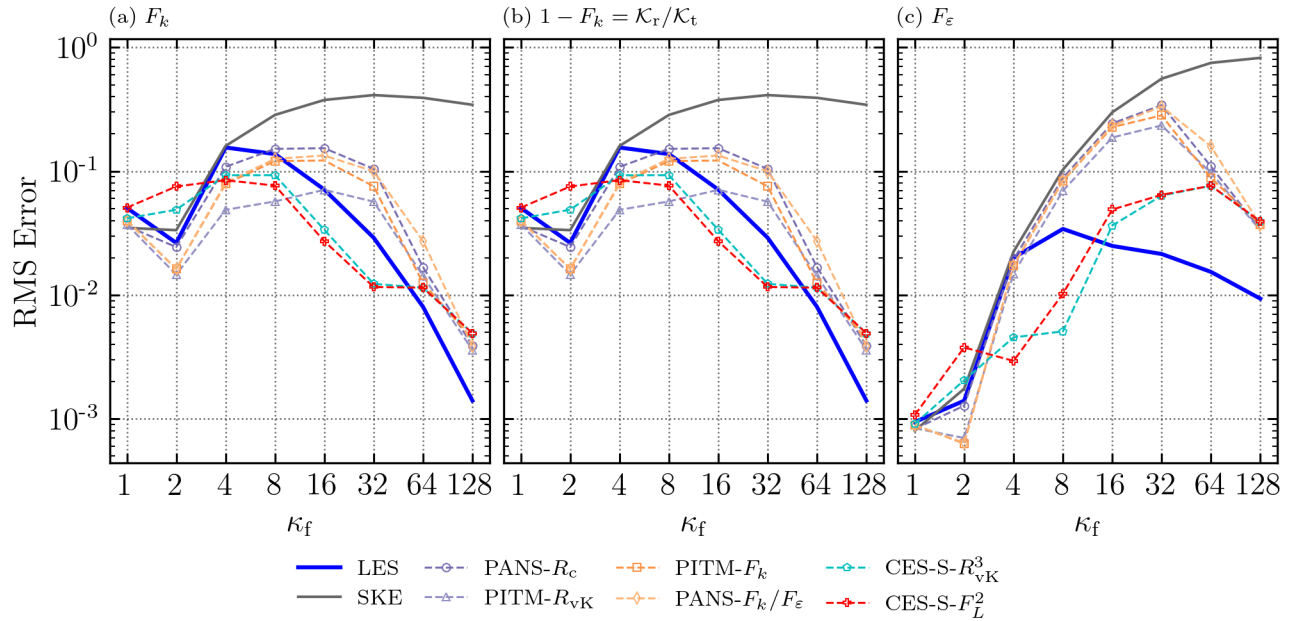


FIG. 1. Root-mean-square error (RMSE) at each filter wavenumber κ_f of (a) modeled KE fraction, F_k , (b) resolved KE fraction, $1 - F_k$, and (c) modeled KE dissipation fraction F_ε for all PANS-like models tested, and in comparison to the k -equation LES model and standard k - ε (SKE) model with constant coefficients.

Alt Text: Root-mean-square error (RMSE) at each filter wavenumber of (a) modeled KE fraction, (b) resolved KE fraction and (c) modeled KE dissipation fraction for all PANS-like models tested, and in comparison to the k -equation LES model and standard SKE model with constant coefficients.

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are clearly more accurate in F_ε compared to the PANS/PITM models from $\kappa_f = 4$ to 32, while being better or the same at $\kappa_f = 64$ and 128. All PANS-like models clearly outperform SKE for $\kappa_f \geq 4$, and are inferior to k -equation LES for $\kappa_f \geq 64$.

In Fig. 2 we present the *relative* error for the same quantities of interest as the absolute logarithmic error (ALE) after interpolating data to 100 geometrically-spaced time points. This relative error corresponds to the mean absolute deviation from unity of each time series as visually displayed in Fig. 3 on log-log axes, and is computed as

$$\text{ALE}(F_x) = \exp\left(\frac{1}{N_t} \sum_{n=1}^{N_t} |\ln [F_x(t'_n)/F_x^{\text{dns}}(t'_n)]|\right) - 1.0 \quad \text{where } \ln(t'_{n+1}/t'_n) = \ln(t'_{\text{end}})/(N_t - 1) \forall t'_n \in [1, t'_{\text{end}}]. \quad (2)$$

In contrast to the RMSE, the ALE of F_k and $1 - F_k$ have differing trends, where $\text{ALE}(F_k)$ is increasing roughly monotonically with κ_f for all PANS-like models while $\text{ALE}(1 - F_k)$ is decreasing roughly monotonically for all PANS-like models. It can be seen that no single model has the lowest $\text{ALE}(F_k)$ across all κ_f , but the CES-S models are best or nearly best from $\kappa_f = 8$ to 32, but degrade in quality relative to PANS/PITM at $\kappa_f \geq 64$. In contrast, the CES-S models are competitive or better than PANS/PITM in $\text{ALE}(1 - F_k)$ for all filter scales except $\kappa_f = 2$. Similarly, CES-S is competitive or better than all PANS/PITM in $\text{ALE}(F_\varepsilon)$ for all filter scales except $\kappa_f = 2$ and 128. As with RMSE, the SKE model shows inferior performance generally for all $\kappa_f \geq 4$, while k -equation LES is clearly better than all PANS-like models for all $\kappa_f \geq 64$.

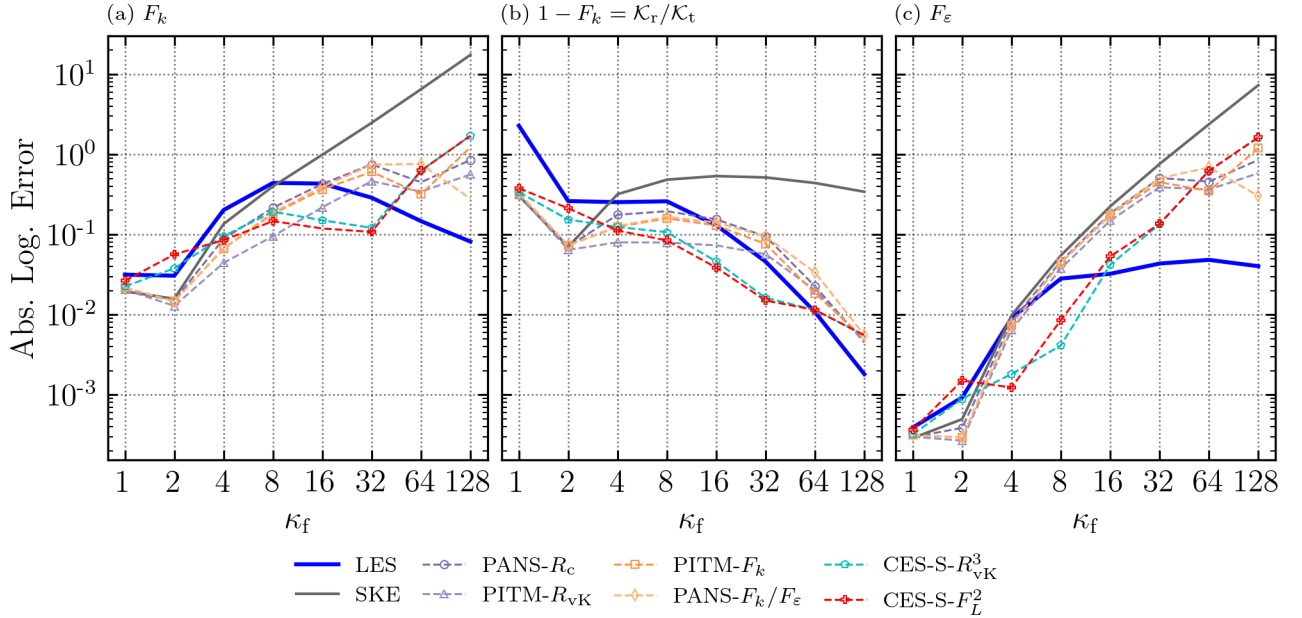


FIG. 2. Mean absolute logarithmic error (ALE) at each filter wavenumber κ_f of (a) modeled KE fraction, F_k , (b) resolved KE fraction, $1 - F_k$, and (c) modeled KE dissipation fraction F_ε for all PANS-like models tested, and in comparison to the k -equation LES model and standard k - ε (SKE) model with constant coefficients.

Alt text: Mean absolute logarithmic error (ALE) at each filter wavenumber of (a) modeled KE fraction, (b) resolved KE fraction, and (c) modeled KE dissipation fraction for all PANS-like models tested, and in comparison to the k -equation LES model and standard SKE model with constant coefficients.

Figure 3 presents time series of the ratios F_x/F_x^{dns} for each of F_k , $1 - F_k$ and F_ε plotted against the normalized decay time t' on log-log axes, which visualizes the ALE integrand at all κ_f . Figure 4 presents time series of the so-called “imposed-actual resolution mismatch”, “actual-true resolution mismatch”, and “imposed-true resolution mismatch” ratios for each model, with the resolution control parameter defined for each model in Table 1.

TABLE I. Summary of turbulence model resolution control parameters.

Alt Text: Summary of control parameters for the turbulence model resolution.

Model	Imposed Model-Input	Actual Model-Output	True (filtered DNS)
PANS- R_c	$R_{cp} = \text{const} = F_k^{\text{dns}}(t=0)$	$R_{\text{actual}} = F_k(t)/F_\varepsilon(t)$	$R_{\text{dns}} = F_k^{\text{dns}}(t)/F_\varepsilon^{\text{dns}}(t)$
PITM- R_{vK}	$R_{cp}(t) = R_{vK}(t)$	$R_{\text{actual}} = F_k(t)$	$R_{\text{dns}} = F_k^{\text{dns}}(t)$
PITM- F_k	$R_{cp}(t) = F_k(t)$	$R_{\text{actual}} = F_k(t)$	$R_{\text{dns}} = F_k^{\text{dns}}(t)$
PANS- F_k/F_ε	$R_{cp}(t) = F_k(t)/F_\varepsilon(t)$	$R_{\text{actual}} = F_k(t)/F_\varepsilon(t)$	$R_{\text{dns}} = F_k^{\text{dns}}(t)/F_\varepsilon^{\text{dns}}(t)$
CES-S- R_{vK}^3	$R_{cp}(t) = R_{vK}^3(t)$	$R_{\text{actual}} = F_L^2(t)$	$R_{\text{dns}} = [F_L^{\text{dns}}]^2(t)$
CES-S- F_L^2	$R_{cp}(t) = F_L^2(t)$	$R_{\text{actual}} = F_L^2(t)$	$R_{\text{dns}} = [F_L^{\text{dns}}]^2(t)$

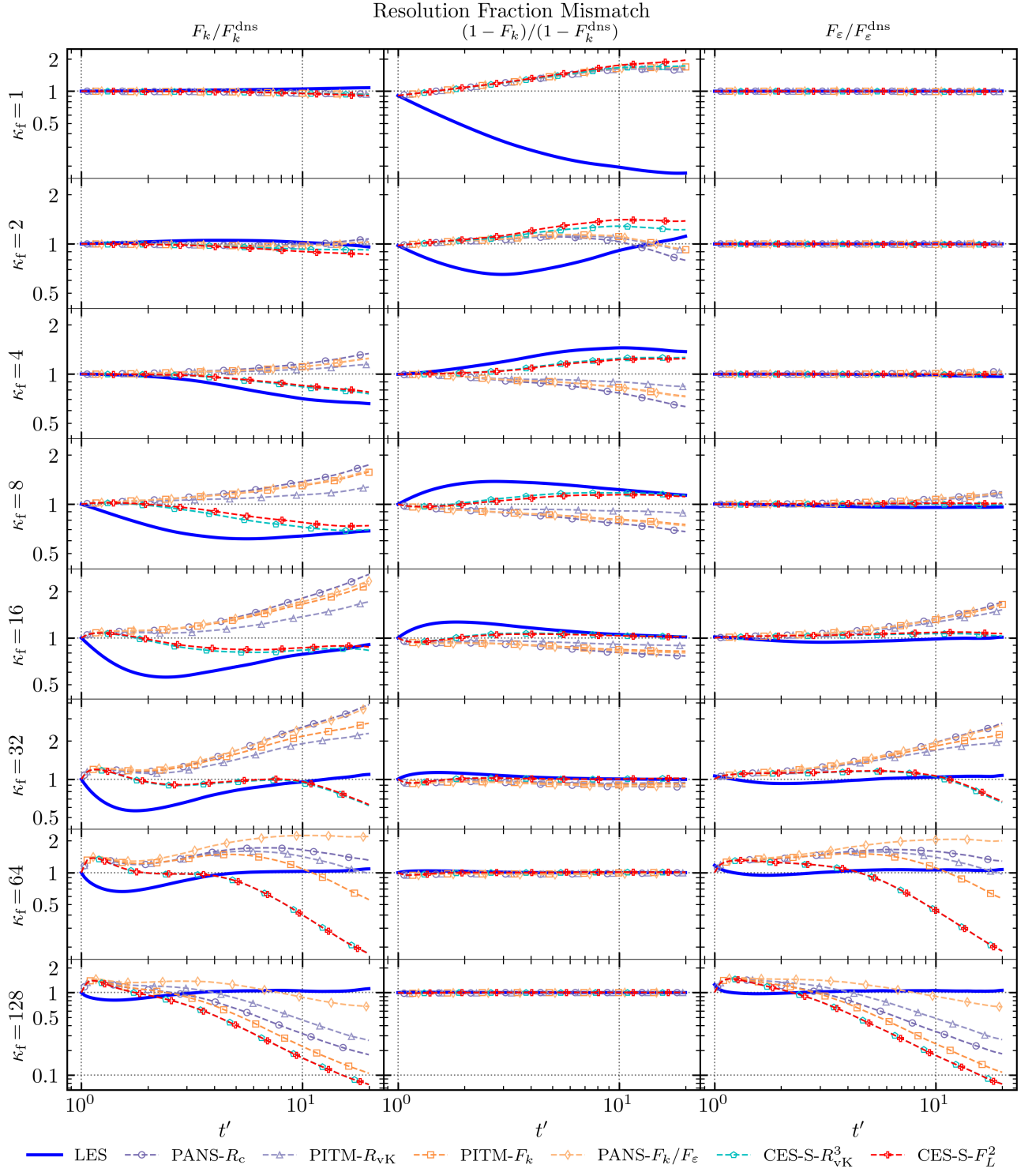


FIG. 3. Time series of the ratios F_x/F_x^{dns} for each of (left) F_k , (center) $1 - F_k$ and (right) F_ϵ plotted against the normalized decay time t' at each filter wavenumber κ_f .

Alt Text: Time series of the resolution fraction for (left) modeled KE fraction, (center) resolved KE fraction, and (right) modeled KE dissipation fraction plotted against the normalized decay time t' at different filter wavenumbers.

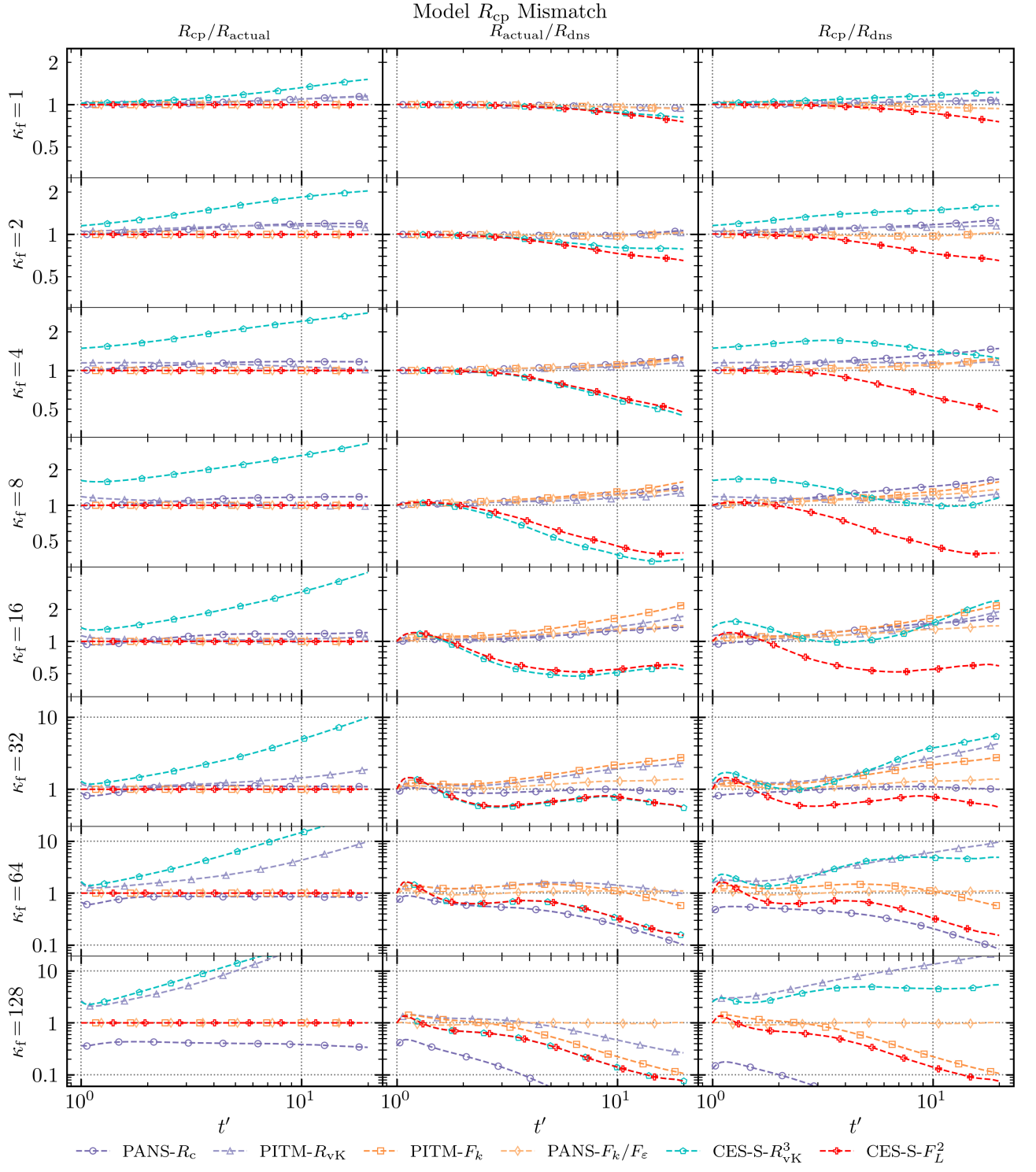


FIG. 4. Time series of the (left) “imposed-actual resolution mismatch”, (center) “actual-true resolution mismatch”, and (right) “imposed-true resolution mismatch” ratios for each model plotted against the normalized decay time t' at each κ_f . See Table 1 for extra details.

Alt Text: Time series of the ratios of imposed to actual resolution mismatch (left column), actual to true resolution mismatch (center column), and imposed to true resolution mismatch (right column) ratios for each model plotted against the normalized decay time t' at different filter wavenumbers.

DATA AVAILABILITY STATEMENT

Raw simulation data were generated on large-scale LANL computing facilities. Derived data supporting the findings of this study can be made available from the authors upon request. The simulation code used to generate all data is available at <https://github.com/lanl/ShenCFD>.

¹S. Heinz, “Comment on ‘Posterior comparison of model dynamics in several hybrid turbulence model forms’ [Phys. Fluids 36, 105148 (2024)],” *Physics of Fluids* (2026).

²C. A. Z. Towery, J. A. Sáenz, and D. Livescu, “Posterior comparison of model dynamics in several hybrid turbulence model forms,” *Physics of Fluids* **36** (2024).